

MUSEUM OF TOLERANCE JERUSALEM

The Sanhedrin Experience

Disagree Better. Discover Together.

Complete Implementation Architecture Version 3.1

Field	Detail
Version	3.1 -- Implementation Architecture (Revised)
Date	March 2026
Authors	Daniel Muller & Jonathan Sagir
Status	Pre-POC / Architecture Definition
Scope	Full system + runtime, safety, resilience, observability
Incorporates	Runtime control model, safety from day 1, degraded modes, ModeProfile config, LLM abstraction, observability, visual/avatar pipeline, WhatsApp POC spec. All approved comment revisions applied.

Museum of Tolerance - Proprietary spec sheet - to be viewed only after NDA.

1. Executive Summary

The Sanhedrin Experience is a multi-agent AI system powering an interactive civic discourse installation at the Museum of Tolerance Jerusalem. Visitors engage with 80 AI-powered historical Jewish figures (Sages) through structured debate on moral dilemmas, facilitated by a human Host and orchestrated by AI meta-agents.

Core Design Principle

Shared spine, swappable persona. One orchestration pipeline, one vector store, one session engine. What makes a Sage a Sage is a database row and a prompt template -- not a separate infrastructure component. Adding a new Sage costs zero infrastructure.

Dual-Mode Operation

The same system supports two fundamentally different cognitive architectures: Culture of Disagreement (pedagogical, process-focused, collaborative) and Debate Mode (competitive, outcome-focused, rhetorical). Every agent -- Nasi, Av Beit Din, and each Sage -- carries distinct behavioral expertise for each mode.

Azure Ecosystem (TBD, might be same level)

Sanhedrin runs in the same Azure tenant as the existing Rambam (Maimonides) exhibit, sharing infrastructure (VNet, Bastion, WAF, Key Vault, DevOps, Azure OpenAI endpoint). Separate resource group, shared spine.

Visual Pipeline

AI-driven talking portraits powered by Azure Speech viseme data, displayed as CGI-grade visual representations of each character. One full-body captured animation per Sage plus real-time face sync from TTS audio. No per-character video production.

Note: The Pepper's ghost/holographic technique inherited from Rambam is under active re-evaluation. A high-resolution display approach may replace it. Decision pending meeting alignment between Daniel and Jonathan.

POC Strategy

A real WhatsApp and/or Telegram group (platform TBD) with 1 human, 2 AI Sages, and 1 AI moderator. Zero UI development. Tests the pure cognitive engine -- the 30% that is unsolved.

Resilience from Day 1

Runtime control model with session ownership and reconnect behavior. Safety layer in every phase. Degraded modes for every failure scenario. Structured observability on every turn. LLM abstraction layer for provider optionality.

2. Strategic Architecture

This section details the Strategic Architecture for the Sanhedrin Experience. This is not a strict, final specification, but a phased framework designed for Knowns/Unknowns Adjustability, recognizing that approximately 70% of the system's infrastructure is defined, while 30% -- the complex, real-time cognitive interaction -- remains a discovery and tuning process. The architecture is explicitly designed as a flexible Scenario Design System to accommodate factors yet to be discovered, rooted in four critical design pillars:

- **Bi-Mode (Debate and Discourse):** The foundational ability to instantly switch the entire system's cognitive framework between the collaborative Culture of Disagreement and the competitive Debate Mode.
- **Swappable Persona and System Integrity:** The Shared Spine principle allows for dynamic management (addition or removal) of all 80+ agent roles via prompt injection, guaranteeing the underlying system integrity is never compromised.
- **Facilitator Dashboard (Monitoring and Navigation):** A dedicated control panel (Host Tablet and Analytics Dashboard) provides the human facilitator with real-time monitoring and override capability, ensuring the Host always has ultimate authority to navigate the audience and participants, and intervene during unforeseen events.
- **Platform Adaptability and Spin-off:** The decoupled architecture (Layers 3, 5, and 7) is built to support a potential EdTech platform spin-off. Key assets like the 80 Sages, the four-layer knowledge architecture, and the personalized takeaway components are intentionally modular for future monetization and educational outreach beyond the museum walls.

2.1 Role Equals Function, Not Staffing

Every component is defined by what it does, not who performs it. The Nasi can be AI, human, or hybrid. Same codebase, all modes. This principle sits within the 70% known layer -- it is a foundational constraint that does not change regardless of what is discovered during development.

2.2 Two-Layer Cognitive Architecture

Layer 1 (Orchestration): Nasi and Av Beit Din use the 13 Thinking Modes as a strategic framework, deciding which mode is needed in real time. Layer 2 (Character): 80 Sage agents use Thinking Modes as personality and cognitive style constraints. The orchestration logic is defined; the emergent behavior of the interaction between layers is part of the 30% discovery process.

2.3 Shared Spine, Swappable Persona

One pipeline, one vector store, one session engine. A Sage is a row in the sages table (biographical embedding, thinking modes, wisdom domain, archetype) plus prompt template injection at runtime. This is the technical backbone of Design Pillar 2 and is fully within the 70% known layer.

2.4 Dual-Mode Cognitive Mastery

Every agent carries two complete behavioral models -- Culture of Disagreement and Debate Mode. The active model is injected per session. Agents do not simply change tone; they change what they consider a good contribution. This is the technical implementation of Design Pillar 1, and how the switch performs under real audience conditions is a primary 30% unknown.

2.5 Four-Layer Knowledge Architecture

Agent expertise is organized in four layers: K1 (Discourse Methodology), K2 (Mode-Specific Expertise), K3 (Thinking Mode and Mode behavioral profiles), K4 (Biographical Persona). All layers are injected via prompt engineering, not fine-tuning. This architecture is the modular foundation for Design Pillar 4 and the future EdTech spin-off potential.

2.6 Context Is King -- Session Context as a First-Class Concern

The session must be formally understood as a complex, evolving context window. The entire flow -- from the Nasi selecting Sages, to the A2A2H Protocol validation steps, to generating coherent arguments and personalized takeaways -- relies on perfect context management. Context management is a dedicated architectural layer to be developed. The complexity of managing this evolving context window, especially during lengthy sessions, is a critical factor in performance and coherence, and sits squarely within the 30% unknown layer.

2.7 Incremental Azure Deployment

Separate resource group, shared VNet, Bastion, WAF, Key Vault, DevOps, and Azure OpenAI endpoint with Rambam. This is a 70% known -- infrastructure decisions are defined, with configuration tuning expected as load and session complexity become clearer.

2.8 Educational, Not Judicial

No winner in Discourse mode, and a structured winner in Debate mode only. The experience succeeds when participants think better. This is the north star principle that governs all design decisions across both the 70% known and 30% unknown layers.

2.9 Privacy-First Design

Anonymization is built into the core. No PII beyond session lifetime. GDPR-compliant. This is a non-negotiable constraint in the 70% known layer and applies equally to the EdTech spin-off scenarios.

2.10 Safety from Day One

Content filtering, topic classification, and transcript logging are present in every phase, including POC. Prompt drift, persona drift, and sensitive-topic failures will surface earliest in development, which is precisely when guardrails must exist. This principle directly serves the 30% discovery process -- instrumented safety is how unknowns are safely explored.

2.11 Graceful Degradation

Every layer has a defined fallback. If avatar fails: static portrait plus audio. If TTS fails: on-screen text. If LLM times out: Host carries. If WebSocket drops: cached state. The experience never stops -- it simplifies. This directly supports Design Pillar 3, ensuring the Facilitator always has a functioning system to navigate.

2.12 Structured Observability

Every turn logs session ID, turn ID, speaker, move type, prompt version, model, latency, and validator flags. Improvement without instrumentation is guesswork. Observability is the primary mechanism for converting the 30% unknown into knowledge; it is not a nice-to-have but a core architectural requirement from day one.

3. Rambam Architectural Lessons

The Rambam exhibit serves as the proven sibling project. Key architectural lessons that inform Sanhedrin.

Core Cognitive Difference: Interaction vs. Structured Scenario

Before detailing what transfers and what does not, it is important to establish the fundamental cognitive difference between the two systems. Rambam operates as a single-character interaction -- one AI persona responding to open-ended visitor input without a predefined argumentative structure. Sanhedrin operates on an entirely different cognitive model: every session is anchored to a structured Scenario Frame -- a specific moral dilemma with defined sides, discourse rules, and a progression arc. This structured session frame is a new, non-inherited element with no equivalent in the Rambam architecture. It requires dedicated development effort and is one of the primary contributors to the 30% unknown layer of this system.

What Sanhedrin Inherits from the Rambam Infrastructure

- Azure Bastion, Application Gateway WAF v2, VNet, Private Endpoints, and DNS zones
- Key Vault, Azure DevOps pipelines and repos, Static IP allocation
- Azure OpenAI endpoint (shared deployment, Sanhedrin workloads added)
- AV control pipeline patterns (trigger, render, display)
- Holographic display hardware and Pepper's ghost technique (under re-evaluation -- see note in Section 1)

What Sanhedrin Explicitly Does Not Replicate Per-Sage

- Azure AI Search indices -- replaced by pgvector in shared PostgreSQL
- Cosmos DB -- replaced by PostgreSQL (already needed for sessions)
- Redis Premium -- replaced by Redis Basic/Standard (right-sized per phase)
- Custom Neural Voice models -- replaced by standard neural voices with SSML prosody tuning
- Per-character infrastructure deployments of any kind

Core Architectural Shift

Rambam = one character, deep infrastructure per persona. Sanhedrin = 80 characters, shared spine with persona injected at the prompt layer. Adding a new Sage requires a database row and a prompt template -- not a new search index, voice model, or infrastructure stack.

4. System Layers

The system is organized into 9 layers, from physical hardware to AI orchestration and post-session output.

#	Layer	Function
0A	Physical & Presentation	12 kiosks, Audience Pathfinder devices, Host panel, lighting, Sage Screens, Host Screen
0B	Visual / Avatar Pipeline	Sage portraits, AI talking-portrait renderer, holographic/high-res display (TBD), Virtual Audience projection
1	Gateway & Sessions	API gateway, session lifecycle, WebSocket, device pairing
2	Orchestration	Nasi (conductor), Av Beit Din (quality control), Mode Router
3	Character Agents	80 AI Sage personas with biographical grounding, Thinking Mode constraints, wisdom domains
4	Infrastructure Agents	Hidden autonomous pre-processing layer. Atomic-task agents fire in parallel on every turn: Sentiment, Fact Detection, Example Generation, Research, Synthesizer, Cultural Sensitivity, Devil's Advocate, Context. Never visible to participants.
5	Engagement & Analytics	Pre, During, and Post session: Reactions, Gold Star, Puzzle, Evaluation Tracker, Polarization Score, L'Shem Shamayim Index, Facilitator Intelligence Dashboard
6	Data & Persistence	Memory and context engineering backbone. Manages context injection (amount, order, cost) and persists sessions and assets. Includes Context Registry.
7	Output & Takeaway	QR personalized summary, teacher materials, analytics dashboard

Layer 0A: Physical and Presentation (Museum)

Component	Description	Tech
12 Speaker Kiosks	Touchscreen stations in semi-circle (curved bench with indentations per Yazdani design)	React Electron app, capacitive touch
Audience Pathfinder Devices	Dedicated MOTJ devices distributed to participants via QR pairing. Named 'Pathfinder' to distinguish from personal phones.	Mobile PWA

Component	Description	Tech
Host Tablet	Facilitator control panel: advance steps, trigger Sages, override selections, launch evaluations	React admin panel on tablet
2 Sage Screens	Large portrait displays flanking Host Screen, show active Sage per Side with AI talking-portrait animation	High-res display + avatar renderer
Host Screen	Central display: Wheel of Sages, puzzle reveal, evaluation stats, round headers	React canvas / WebGL
Dynamic Lighting	Individual spotlights on 12 Speaker positions, colored per Side, brightens for active Speaker	DMX / ArtNet integration
Virtual Audience	Projected crowd on walls that reacts to engagement signals (puzzle pieces, reaction spikes)	Pre-built reaction animation loops

Layer 0B: Visual / Avatar Pipeline

The visual representation of Sages is a critical experience layer completely absent from the Rambam-multiplied model. The pipeline scales to 80 characters without per-character video production.

Avatar Technology: AI-Driven Talking Portraits (Recommended)

One high-resolution portrait image per Sage plus real-time face animation driven by the TTS audio stream. Azure Speech outputs viseme data (blend shape weights at 60fps) alongside audio. This feeds directly into portrait animation renderers (D-ID, HeyGen, Azure Talking Avatar, or SadTalker).

Phase	Visual Capability	Implementation
POC	Text only. Sage identity via emoji + name prefix in WhatsApp.	No visual pipeline needed.
Demo	Static portraits displayed during speech + TTS audio. AI talking-portrait prototype for 2-3 key Sages.	Portrait images + Azure Speech TTS + D-ID/HeyGen API for animation.

Phase	Visual Capability	Implementation
Museum	Full AI talking portraits on 2 Sage Screens. Dynamic character switching (Wheel of Sages: portrait loads in under 2 seconds). Spotlight sync. Virtual Audience reactions.	Portrait renderer on local GPU. Azure Speech viseme stream. DMX lighting sync via WebSocket.

Portrait States per Sage

State	Visual Behavior	Trigger
Idle	Subtle breathing/movement loop. Eyes open, attentive.	Sage is assigned to Side but not currently speaking.
Speaking	Lip-sync to TTS audio (viseme-driven). Gesture emphasis on key words. Gaze toward audience.	Sage's turn. Audio stream active.
Listening	Attentive stillness. Slight head movement. Occasional blink.	Other Sage or human is speaking.
Reacting	Brief expression change: raised eyebrow (challenged), nod (agrees), slight lean forward (intrigued).	Triggered by Nasi or engagement signals.
Entering	Fade-in / walk-in animation when selected via Wheel of Sages.	Wheel stops on this Sage.

Rambam Hologram to Sanhedrin: What Transfers

- Display hardware (Pepper's ghost or high-res alternative -- under evaluation) transfers directly
- AV control pipeline (trigger, render, display) transfers with modification for dynamic character switching
- Custom Neural Voice hosting (\$605/mo per model) does NOT transfer -- replaced by shared standard neural TTS
- Pre-recorded video segments do NOT transfer -- replaced by real-time AI portrait animation

Key requirement: Character switching in under 2 seconds. Rambam never switches characters. Sanhedrin switches every time the Wheel of Sages spins or a new Side speaks.

Layer 1: Gateway and Session Management

Component	POC	Dem o	Museu m	Notes
WAF / App Gateway	--	--	Shared	Already paid by Rambam. Add backend pool.

Component	POC	Dem o	Museu m	Notes
API Management	--	--	New	Consumption tier, ~\$100/mo
Session Manager	Yes	Yes	Yes	Node.js. State machine: CREATE > SETUP > DILEMMA > EVAL > ROUND_1 > MID > ROUND_2 > FINAL > SYNTHESIS > ARCHIVED
WebSocket Server	Yes	Yes	Yes	Socket.io. Real-time bidirectional communication across all devices.
Device Pairing	--	--	Yes	Kiosk + MOTJ Pathfinder device association via QR.
Host Tablet API	--	Yes	Yes	Endpoints: advance_step, trigger_sage, override_speaker, trigger_puzzle_piece, launch_evaluation, present_results.

Layer 2: Orchestration (Meta-Agents)

Two meta-agents that do NOT generate visitor-facing dialogue -- they route, decide, and intervene. Both carry dual-mode expertise.

2.1 Nasi (Conductor)

LLM: Primary Model Tier (GPT-4 class or equivalent, managed via LLM Abstraction Layer) | Thinking Modes: #1, #2, #4, #10, #13

Responsibilities:

- Select 5-7 Sages from the 80-agent pool based on dilemma relevance, mode diversity, era diversity, and a wildcard.
- Manage the Wheel of Sages mechanic: pre-filter eligible Sages per Side, populate the wheel, controlled reveal.
- Decide speaking order each turn: alternate sides, enforce Thinking Mode diversity, invite humans every 2-3 exchanges.
- Call for specific discourse move types: Kushiya, Terutz, Bridge, Reframe, Story, Core Question.
- Intervene on: circular argument (3+ turns, no new claims), one side dominating (>70%), factual claim needs verification, time running out.
- Manage Sage fallback: if a human Speaker fails to communicate, trigger the Sage on that Side to present the argument instead.

Aspect	Culture of Disagreement	Debate Mode
Pacing	Slower. Allow pauses. 'Take a moment to consider.'	Faster. Time pressure. 'You have 30 seconds.'
Turn allocation	Equal airtime. Invite human every 2nd exchange. Redirect if one side >60%.	Structured rounds with timed rebuttals. Cross-examination exchanges.

Aspect	Culture of Disagreement	Debate Mode
Preferred moves	Bridge, Story, Core Question, Reframe	Challenge (Kushiya), Attack, Counter (Terutz), Evidence demand
Intervention	Consensus forming too early: Devil's Advocate. Energy dropping: Story.	Weak argument: demand evidence. Evasion: demand direct answer.
Closing	Synthesize themes. 'What's one thing from the other side?' No winner.	Audience votes. Gold Star. Nasi summarizes strongest arguments per side.

2.2 Av Beit Din (Quality Control)

LLM: Primary Model Tier | Thinking Modes: #6 (Skeptical-Critical), #4 (Active), #13 (Collective)

Enforces Sanhedrin Rules of Engagement:

- State the opponent's case first before presenting your own.
- Attack the premise, not the person.
- Concede the gray area -- acknowledge what the other side gets right.
- Ground arguments in practice, not just theory.

POC: Not active. Quality enforced via Sage prompt constraints. Demo: Active, validates outputs pre-delivery. Museum: Full authority + human-escalation for cultural sensitivity.

2.3 Mode Router

Aspect	Culture of Disagreement	Debate Mode
Goal	Process exploration. Expand understanding.	Persuasion and rhetoric. Best argument wins.
Sage behavior	Build on opponent's arguments. Find merit. Hedge and question.	Advocate fiercely. Target weakest link. Decisive language.
Scoring	L'Shem Shamayim Index + Polarization Score	Audience votes per round + final vote
Target audience	School groups, youth movements	General visitors, corporate groups

Layer 3: Character Agent Layer (80 Sages)

Core insight: A Sage is a prompt injection, not an infrastructure component. Each Sage is a row in PostgreSQL containing biographical embedding, thinking mode constraints, wisdom domain, archetype, signature rhetoric, and voice config.

The 13 Talmudic Thinking Modes

#	Mode	Cognitive Instruction
1	Disruption	Challenge assumptions. 'What if the opposite is true?'

#	Mode	Cognitive Instruction
2	Flexible	Adapt approach based on what the discussion needs now.
3	Narrative-Visual	Ground argument in a specific story or lived experience.
4	Active	Push for action. 'What do we DO about this?'
5	Interpretive	Find deeper meaning. 'What's really being said underneath?'
6	Skeptical-Critical	Demand evidence. 'On what basis do you claim this?'
7	Generative	Create new possibilities. 'What if we considered...'
8	Imaginative	Propose thought experiments. 'Imagine that...'
9	Creative	Connect disparate ideas. 'This reminds me of...'
10	Solution-Oriented	Focus on practical outcomes. 'Here is what the law should be...'
11	Innovation-Focused	Seek systemic redesign. 'The whole framework needs rethinking...'
12	Entrepreneurial	Ask who does the work. 'Who builds this? Who pays?'
13	Collective	Speak for the group. 'What does this mean for all of us?'

Discourse Move Taxonomy

Move	Hebrew	When Used / Sage Constraint
Opening argument	Taanah	Start of round. Present your side's core position with reasoning.
Challenge	Kushiya	After opposing argument. Find the specific weakness in what was just said.
Response	Terutz	After being challenged. Defend with concrete counter-evidence or principle.
Scenario	Okimta	Test a general rule with a specific case that exposes its limits.
Reframe	Ipcha Mistabra	When stuck. Argue the OPPOSITE of your position to test conviction.
Bridge	Shakla v'Tarya	Mid-discourse. Find what's valid in the other side's argument.
Switch	Switch	Argue FOR the other side to demonstrate understanding.
Core question	B'mai Ka Miflagi	Nearing synthesis. Name the fundamental disagreement underneath.
Story	Maaseh	End of round. Share a concrete story from your life/era that illuminates the dilemma.
Information	Meida	Inject facts, data, or historical context. Grounded in sources.

Layer 4: Infrastructure Agents (Hidden Pre-Processing Layer)

Layer 4 operates as a hidden autonomous pre-processing layer that runs beneath every single turn of the discourse. These agents are never visible to participants -- they fire in parallel, complete atomic tasks, and feed structured signals back to the Nasi and Av Beit Din before any response reaches the surface.

The distinction from character and orchestration layers is intentional: these are not conversational agents. They do not speak, argue, or decide. They detect, classify, score, and inject -- each one responsible for exactly one atomic function.

LLM Tier: Lightweight Model Tier (classification and validation tasks -- specific model binding managed via LLM Abstraction Layer, Section 9.3).

Agent	Atomic Function	POC	Dem o	Museu m
Sentiment Agent	Tone and emotional register scoring per turn	--	Yes	Yes
Fact Detection Agent	Flags factual claims requiring verification before delivery	--	Yes	Yes
Example Generation Agent	Surfaces relevant historical examples or analogies available for Sage use	--	--	Yes
Research Agent	Real-time data and source injection	--	Yes	Yes
Synthesizer	Theme aggregation, Polarization Score computation	--	Yes	Yes
Cultural Sensitivity Agent	MOTJ values alignment, flags escalation triggers	--	--	Yes
Devil's Advocate Agent	Groupthink detection, minority perspective surfacing	--	--	Yes
Context Agent	Selective context window management across all active agents	--	--	Yes

Design principle: Each agent owns one atomic task and does it completely. No agent in this layer makes narrative or conversational decisions -- that authority belongs exclusively to the Nasi (Layer 2). Layer 4 agents inform; they do not decide.

Layer 5: Engagement and Analytics -- Pre, During, and Post Session

Layer 5 serves three distinct temporal functions. Before the session it establishes baselines. During the session it gives the facilitator a live, data-driven decision-making instrument. After the session it generates insights for participants, educators, and operators. The Host should never have to guess whether the room is engaged, polarized, shifting, or losing energy -- Layer 5 makes all of that visible in real time.

Pre-Session

Component	POC	Dem o	Museu m	Description
Evaluation Tracker (Baseline)	Yes	Yes	Yes	Initial position + confidence captured before discourse begins. Scale 1-5. Establishes the baseline against which all shift is measured.
Dilemma Briefing Display	--	Yes	Yes	Participants receive the scenario frame and side assignments before the first turn. Sets cognitive context.

During Session -- Facilitator Intelligence Dashboard (Host Tablet Live View)

This is the nerve center of Host decision-making. Every engagement signal feeds into a live view visible only to the facilitator, enabling intervention before problems surface rather than after.

Signal	What It Shows	Host Action Enabled
Live Polarization Score	Group divergence between sides, updating every turn	Inject Bridge move, invite swing participant, adjust pacing
Engagement Heatmap	Which participants are reacting, voting, or going silent	Target silent participants, redirect discourse energy
Argument Momentum	Which side is gaining traction turn by turn	Balance airtime, trigger Devil's Advocate agent
Position Shift Tracker	Individual confidence changes from baseline in real time	Identify minds in motion, elevate their voice
Discourse Move Distribution	Move types used vs. missing in current session	Call for Story, Reframe, or Core Question to diversify
L'Shem Shamayim Live Score	Real-time quality index per Sage and per side	Flag low-quality exchanges, prompt Av Beit Din intervention
Alert Feed	Flagged turns: persona drift, sensitive topic, one side dominating	Direct intervention, escalation, or emergency stop

During Session -- Participant Engagement Components

Component	POC	Dem o	Museu m	Description
Evaluation Tracker (Midpoint)	Yes	Yes	Yes	Position + confidence at session midpoint. Delta from baseline visible to Host.
Reaction Buttons	--	Yes	Yes	Thumbs Up (I agree) + Question Mark (makes me think). Both positive signals.
Gold Star Award	--	--	Yes	Audience awards 1 star to most valuable Speaker. Movable until final evaluation.

Component	POC	Dem o	Museu m	Description
Puzzle Reveal	--	--	Yes	Visual on Host Screen fills as arguments reveal truths. Auto + Host-triggered.
L'Shem Shamayim Index	--	Yes	Yes	Culture mode only. Composite: acknowledgment of other side, reasoning quality, position shift, rules compliance.
Polarization Score	--	Yes	Yes	Culture mode. Group divergence over time. Decreasing = success.

Post-Session

Component	POC	Dem o	Museu m	Description
Evaluation Tracker (Final)	Yes	Yes	Yes	Final position + confidence. Full delta from baseline to close. Primary impact measurement.
Session Summary	--	Yes	Yes	Aggregate position shift, top arguments, discourse move distribution, engagement score. Operator-facing.
Participant QR Takeaway	--	--	Yes	Personalized summary: starting position, what shifted, confidence trajectory.
Analytics Export	--	Yes	Yes	Full session data export for Bar-Ilan research store and MOTJ operator dashboard.

Layer 6: Data, Persistence, and Context Engineering

Layer 6 is not simply storage -- it is the memory and context engineering backbone of the entire system. Beyond persisting sessions and assets, it is responsible for managing what each agent knows at any given moment: which context is injected, how much, in what order, and at what cost. Context engineering is a first-class architectural concern here, not an afterthought.

Context Engineering Management

Every agent operates within a finite context window. Layer 6 owns the strategy for filling that window intelligently on every turn:

- Prompt prefix caching: the K1-K4 knowledge layers are cached at the infrastructure level, reducing per-turn token cost by an estimated 70-75% of input tokens.
- Context compression: older turns are summarized and compressed rather than dropped, preserving argumentative continuity without exceeding window limits.
- Selective injection: the Context Agent (Layer 4) pulls only the turns, facts, and biographical fragments relevant to the current discourse move and Sage, not the full session history.

- Embedding-based retrieval: pgvector enables semantic search across the dilemma library, Sage biography corpus, and prior session transcripts to surface the most relevant grounding material per turn.
- Window budget tracking: each agent role has a defined token budget. Nasi, Sage, and Av Beit Din operate on different budgets reflecting their function and cost tier.

Storage Architecture

Store	Purpose	Azure Service
PostgreSQL + pgvector	Sessions, turns, evaluations, 80 Sage bios + embeddings, dilemma library, compressed context snapshots	Azure Database for PostgreSQL Flexible Server
Redis	Session state cache, real-time pub/sub, engagement counters, active context window state per agent	Azure Cache for Redis (Basic to Standard per phase)
Blob Storage	Transcripts, portrait images, research exports, prompt version archives	Azure Blob Storage
Research Store	Anonymized aggregates for Bar-Ilan academic partnership	Same PostgreSQL, separate schema. One-way anonymization.
Context Registry	Prompt version tracking, K1-K4 cache keys, per-agent window budget log. Feeds directly into TurnTrace schema (Section 9.4).	Same PostgreSQL, separate schema.

Layer 7: Output and Takeaway (Museum)

Component	Description
QR Takeaway	Personalized digital summary via QR code. Captures the participant's journey: which side they started on, key arguments that shifted their thinking, their confidence trajectory.
Summary Generator	AI-generated reflection. 3 key insights from the discourse, 1 reflection question, 1 memorable Sage quote. Tailored to the participant's contributions.
Teacher Materials	Post-visit lesson plans, discussion guides for continued classroom engagement. Includes the dilemma, key arguments from both sides, suggested follow-up activities.
Analytics Dashboard	Operator view: session metrics, engagement heatmaps, system health, popular dilemmas, average position shift by topic.

5. Knowledge and Expertise Architecture

Every agent in the system carries four layers of injected knowledge. This is context engineering, not fine-tuning -- the expertise lives in the prompt, not model weights.

Layer	Name	Content
K1	Discourse Methodology	Talmudic discourse framework (machloket l'shem shamayim). Discourse move vocabulary (Kushiya, Terutz, Okimta, etc.). Sanhedrin Rules of Engagement. Same for ALL agents, ALL modes.
K2	Mode-Specific Expertise	What makes a 'good' contribution in each mode. Culture: build on opponent, acknowledge gray, end with questions. Debate: attack weakest link, provide evidence, decisive language. 2 behavioral profiles.
K3	Thinking Mode x Mode	Each of the 13 Thinking Modes has distinct behavioral expertise per operational mode. #1 Disruption in Culture = expand (open new angles). #1 Disruption in Debate = destabilize (undermine foundations). 13 modes x 2 modes = 26 behavioral profiles.
K4	Biographical Persona	Jewish Lives grounding. Archetype title, historical values, signature rhetoric, works referenced. Same across both modes. 80 persona profiles.

Prompt Assembly (per Sage, per turn)

- K1: Discourse Methodology (constant, all agents)
- K2: Mode Expertise (Culture OR Debate)
- K3: Thinking Mode #N expertise for this mode
- K4: Biographical persona (Maimonides, Spinoza, etc.)
- Conversation context (managed by Layer 6 Context Engineering)
- Move type instruction from Nasi

= Complete Sage prompt for this turn

Content authoring requirement: 26 Thinking Mode x Mode behavioral profiles (K3) + 2 mode expertise documents (K2) + 80 biographical personas (K4) + 1 discourse methodology document (K1). The K3 profiles are the critical-path content work -- they need Micha Goodman and Miri Dayan.

Fine-tuning option: If prompt injection alone does not achieve the quality bar, fine-tuning on curated expert-rated transcripts becomes an option at the Demo phase. Start with prompts -- faster, cheaper, fully reversible.

6. Dual-Mode Configuration (ModeProfile)

Both operational modes are governed by a single configuration object injected into every agent. A mode switch is a single-knob change, not scattered prompt edits.

Parameter	Culture of Disagreement	Debate Mode
goals	Expand understanding. Surface complexity.	Sharpen arguments. Test conviction.
turn_time_pressure	false -- allow pauses for reflection	true -- 30-second response windows
preferred_moves	Bridge, Story, Core Question, Reframe	Kushiya, Attack, Terutz, Evidence demand
acknowledgment_required	true (must cite opponent first)	false
hedging_language	true ('I would argue...')	false ('This is clear...')
end_with_question	true (deepens inquiry)	false (ends with challenge)
scoring_method	L'Shem Shamayim Index	Audience votes
winner_declared	false -- synthesis, no victor	true -- audience vote declares winner
closing_ritual	'What did you learn from the other side?'	Gold Star + Nasi editorial summary

All behavioral differences derive from this object. Nasi, Av Beit Din, Sage prompts, and Engagement layer all read the active ModeProfile. One config swap, not a codebase search.

7. The A2A2H Protocol

Agent-to-Agent-to-Human -- the core innovation. AI agents coordinate with each other BEFORE output reaches human participants.

Protocol Flow

- TRIGGER: Nasi decides which Sage speaks next and which discourse move type.
- DRAFT: Primary Model Tier generates candidate response with K1-K4 layers + move constraint.
- VALIDATE (Demo+, parallel): Av Beit Din (quality + rules), Sentiment Agent (tone), Research Agent (facts).
- REVISE: If flagged, regenerate with specific constraints from validators.
- VOICE (Demo+): Azure Speech TTS with Sage-specific SSML prosody config.
- VISUAL (Museum): Viseme stream to AI talking-portrait renderer to Sage Screen.
- DELIVER: Text + audio + visual via WebSocket to all connected clients simultaneously.

Latency Budget

Step	Target	Strategy
Sage draft	1.5-2.5s	Primary Model Tier with cached prefix (75% of input). 3-sentence max. Streaming.
Validation	0.5-1s	Lightweight Model Tier parallel calls. Pre-compiled rule checks.
TTS + viseme	1-2s	Azure Speech streaming. Audio + viseme data in single stream.
Portrait animation	<0.5s	Local GPU. Viseme to face animation is real-time.
TOTAL	< 5s	User sees text immediately (streaming). Audio plays 1-2s later. Portrait animates in sync with audio.

8. Runtime Control Model

Defines who owns what at runtime, how events flow, and what happens when components disagree.

8.1 Session Ownership

The Session Manager is the single source of truth. All state changes flow through it. If the Host tablet, kiosk, audience device, and Nasi AI disagree about current phase or speaker, the Session Manager's state wins. Clients sync via WebSocket, not from each other.

8.2 Host Override Precedence

The human Host always has ultimate authority. Override precedence (highest to lowest):

- Host tablet command (advance, trigger, override, emergency stop)
- Nasi AI decision (next speaker, move type, intervention)
- Automated timers (countdown expiry, auto-advance)
- Participant input (votes, reactions, volunteering)

If the Host advances while the Nasi is mid-decision, the Host wins. The Nasi's pending action is cancelled. The human facilitator is never blocked by AI.

8.3 Command vs Event

Type	Definition	Examples
Command	Intent to change state. Requires validation. May be rejected.	host:advance_step, host:trigger_sage, speaker:submit_vote, nasi:select_speaker
Event	Fact that something happened. Immutable. Broadcast to all.	session:phase_changed, sage:response_delivered, evaluation:submitted, reaction:pressed

8.4 Reconnect and Resume

- Kiosk WebSocket drop: reconnects, receives full state from Redis cache.
- Host tablet disconnect >30s: Nasi enters safe auto-mode (alternates sides, holds at next evaluation checkpoint).
- Audience Pathfinder loses connection: rejoins at current phase. Missed reactions lost; evaluation can still be submitted.
- Redis failure: Session Manager falls back to in-memory state. Active sessions continue; new sessions blocked.

8.5 Timeout and Retry

Operation	Timeout	Fallback
Sage LLM response	5s	Show 'Sage is contemplating...' At 10s, Nasi advances to next speaker.
Validation pipeline	2s	Deliver Sage response unvalidated. Log the skip.
TTS generation	3s	Display text on Sage Screen with static portrait. No audio.
Avatar animation	1s	Static portrait + audio. No lip-sync.
Speaker vote	30-45s	Countdown expires. Sage presents argument directly (fallback).
Human (POC)	180s	Nasi re-engages or lets Sages continue.

9. Resilience, Safety, and Observability

9.1 Safety and Content Moderation

Safety exists in every phase. Prompt drift, persona drift, and sensitive-topic failures appear earliest in development -- exactly when guardrails must exist.

Layer	POC	Demo	Museum
Content filter	Lightweight Model Tier check on every Sage response: on-topic? appropriate? no attacks?	Same + Av Beit Din validates discourse rules	Same + Cultural Sensitivity Agent + MOTJ values
Topic bounds	Blocked topic list per dilemma	Expanded boundaries for 3 dilemmas	Full topic model + dynamic edge-case handling
Transcript log	Every message with full metadata	Same + validator flags + interventions	Same + anonymized research export
Escalation	Manual post-session review	Real-time flag to Host tablet	Host escalation + auto-muting on repeated violations

Layer	POC	Demo	Museum
Persona boundary	Prompt constraint: never break character	Same + Av Beit Din catches drift	Same + periodic audit from transcripts

Blocked Move Types

- No Attack moves on first exchange (must open with claims before challenges).
- No Reframe in final 2 turns (risks confusing synthesis).
- No Story moves from Sages without relevant biographical context (prevents fabrication).
- Culture mode: no consecutive Attack moves from same Side (prevents pile-on).

9.2 Degraded Modes

Every layer has a defined fallback. The experience never stops -- it simplifies.

Failure	Degraded Behavior	Visitor Impact
LLM timeout >5s	'Sage is contemplating...' At 10s, Nasi advances.	Brief pause. Host fills with commentary.
TTS failure	Text on screen + static portrait.	Sage 'speaks' in text. Host reads aloud.
Avatar failure	Static portrait + audio.	Portrait like a painting. Audio works.
WebSocket drop (kiosk)	Reconnect from Redis cache.	Momentary freeze on that kiosk.
Host tablet disconnect	Nasi safe auto-mode. Holds at next eval checkpoint.	Session continues in reduced mode.
Full Azure OpenAI outage	Host manual facilitation. Pre-built fallback content.	Human-only discourse exercise. Sages 'silent today.'

9.3 LLM Abstraction Layer

Thin service owning model selection, retries, rate limiting, provider routing. No direct Azure OpenAI calls from application code. All model bindings are managed here -- specific model names (e.g., GPT-4.1, GPT-4.1-nano) are configuration values, not hardcoded references. This supports the 30% unknown: as models evolve and deprecate over the 1-3 year horizon, the architecture remains stable.

Agent Role	Model Tier	Current Binding	Fallback
Nasi / Sage / Av Beit Din	Primary	GPT-4 class / Primary Tier (configurable)	Smaller Primary Tier model
Infrastructure agents	Lightweight	Lightweight Tier (configurable)	Lightweight Tier (retry)

Agent Role	Model Tier	Current Binding	Fallback
Content filter	Fast	Lightweight Tier (configurable)	Keyword blocklist (no LLM)

Provider swap = config change. If Claude, Gemini, or a future model becomes available, update the model registry. No application code changes. Costs approximately 1 day to build.

9.4 Observability and Tracing

Every turn generates a structured trace record. This is not optional -- it is the data that enables systematic improvement.

KPI	Definition and Target
Persona consistency	% of Sage turns rated in-character by blind reviewer. Target: >90%.
Argument coherence	% of turns referencing specific prior claims. Target: >80%.
Mode separability	Blind reviewer correctly identifies Culture vs Debate from transcript. Target: >85%.
Discourse move variety	Distinct move types used per session. Target: 5+ of 10.
Human influence	Sage responses directly referencing human's argument. Target: 3+ per session.
Position exploration	Participants whose confidence changes initial to final. Target: >30%.
Mean response latency	Nasi trigger to delivery. Target: <3s (POC), <5s (museum).

10. Voice Strategy

Do NOT create custom neural voice models. Use Azure Speech standard neural voices with SSML prosody tuning per Sage archetype.

Archetype	Azure Voice	SSML Tuning
Ancient authority (Moses, Maimonides)	en-US-DavisNeural	Rate 0.85, pitch -5%, deliberate pauses, 'grave' style
Passionate advocate (Goldman, Herzl)	en-US-JennyNeural	Rate 1.1, emphasis on key words, shorter pauses, 'empathetic' style
Reflective witness (Wiesel, Anne Frank)	en-US-AriaNeural	Moderate rate, longer pauses, softer volume, 'sad' style
Analytical mind (Einstein, Spinoza)	en-US-GuyNeural	Measured rate, precise articulation, minimal emotion, 'newscast' style
Collective voice (Golda, Ruth)	en-US-SaraNeural	Warm tone, moderate rate, inclusive emphasis, 'friendly' style

Standard neural TTS with SSML prosody tuning provides voice differentiation across all 80 Sages using just 5 base voice profiles. No custom neural voice model hosting required.

11. POC: WhatsApp / Telegram Group

A real messaging group (WhatsApp and/or Telegram, platform TBD) with 1 human, 2 AI Sages, and 1 AI moderator (Nasi). Zero UI development. Tests the pure A2A2H cognitive engine -- the 30% that is unsolved.

Participant	Identity	LLM Tier
Nasi	Moderator. Opens dilemma, assigns sides, manages turns, calls discourse moves.	Lightweight Model Tier
Sage A	Maimonides (#10 Solution-Oriented). The Supreme Systematizer.	Primary Model Tier (with K1-K4 prompt layers)
Sage B	Spinoza (#1 Disruption). Freedom's Messiah.	Primary Model Tier (with K1-K4 prompt layers)
Human	Daniel, Jonathan, Micha Goodman, or test participant.	--

What the POC Proves

- Can 2 AI Sages maintain distinct personas in the same conversation?
- Can Sages respond to each other's specific arguments?
- Can a human influence the discourse direction?
- Does the Nasi manage pacing and discourse move types effectively?
- Do the dual-mode behavioral constraints produce meaningfully different conversations?

Success Criteria

Criterion	Target
Persona consistency	90%+ turns in-character across both modes
Argument coherence	80%+ turns reference prior arguments specifically
Mode differentiation	Blind reviewer can distinguish Culture vs Debate transcripts
A2A quality	At least 3 direct Sage-to-Sage exchanges per session
Human engagement	Human contributes 4+ substantive messages per session
Latency	<3s per Sage response (text only in POC)

Build Timeline

Day	Deliverable
Day 1	Azure OpenAI deployment (Primary + Lightweight tiers). whatsapp-web.js setup. QR scan. Group created.
Day 2	Orchestrator: session state machine + Nasi logic + Sage prompt assembly (K1-K4) + message routing.
Day 3	First real session. Daniel + Jonathan in the group. Run both Culture and Debate mode. Review transcripts.
Day 4+	Iterate prompts. Try 3rd Sage (Golda Meir). Invite Micha Goodman to evaluate. Test with someone who does not know it is AI.

12. Roadmap

Period	Milestone
Now -- Mar 2026	Architecture definition (this document). Internal presentation. Stakeholder alignment.
Apr -- May 2026	Donor presentations (per-donor customization). RFP preparation. Vendor selection. K1-K4 content authoring begins.
Jun -- Aug 2026	POC: Messaging group, 3 AI agents + 2 humans. Both modes tested. Micha Goodman validation.
Sep -- Nov 2026	Working Demo: Custom React UI. 5 Sages + Nasi + Av Beit Din. Voice (TTS+SSML) for 2-3 Sages. AI talking-portrait prototype. 3 dilemmas. Both modes. Evaluation tracker + L'Shem Shamayim Index.
Dec 2026 -- May 2027	Full prototype: 80 Sages, physical space integration (Sage Screens, lighting, kiosks), holographic/high-res display (TBD), Wheel of Sages, all engagement systems, teacher materials, AV testing, staff training, museum deployment.